Module 4

Computer System Hardware

Computer Science Department · School of ICT · Copperbelt University

Dr. Z Lifelo

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Module 4 — Lecture Overview

Six sections — from hardware components and evaluation to health and responsible disposal:

A — **Hardware Overview — Components & Buying Decisions**

Hardware defined · 5 core components · Factors when purchasing · System requirements · Form factors

B — **Inside the Case — Motherboard, CPU, Memory**

Motherboard · Processor cores · RAM vs ROM · Types of RAM · Virtual memory · Machine cycle · Registers

C — **Input and Output Devices**

Scanners · Printer types (6) · Voice synthesiser · Projectors · Install and evaluate hardware

D — **Storage Solutions — Internal, External & Cloud**

HDD vs SSD · Optical media · USB flash drives · Cloud storage · Choosing the right solution

E — **Network Hardware**

Nodes · Hub · Switch · Router · Modem · NIC · Wi-Fi security risks

F — **Hardware Security, Health & Responsible Disposal**

Theft prevention · Power protection · RSI & ergonomics · Cyberbullying · E-waste & SEM

MODULE 4 · SECTION A

Hardware Overview — Components & Buying Decisions

Hardware defined · 5 core components · Purchase factors · System requirements · Form factors

What is Computer Hardware?

Hardware: The physical device itself and ALL its components — wires, cases, switches, chips, and electronic circuits. You can touch hardware. Software is the instruction set that runs ON hardware. (Campbell et al., 2023)

Components of Computer Hardware:

🧠 Memory

Holds data and programs as they are processed by the CPU. Includes RAM (volatile working storage) and ROM (permanent firmware storage). The OS and apps load into memory to run.

💾 Storage Devices

Long-term data retention — survives power-off. Hard disk drives (HDD), solid-state drives (SSD), optical media (CD/DVD/Blu-ray), USB flash drives, and cloud storage.

⚙ CPU (Processor)

The 'brain' that executes instructions. Performs arithmetic, logic, and control operations via a repeating four-step Machine Cycle: Fetch → Decode → Execute → Store.

⌨ Input & Output Devices

Input: keyboard, mouse, scanner, camera, microphone, barcode reader.
Output: monitor, printer, speakers, voice synthesiser, projector.

📡 Communication Devices

Enable networking and internet access: NIC (Network Interface Card), Wi-Fi adapter, modem, router, switch. Allow sharing of hardware, software, data, and information.

Selecting the Right Computer — Purchase Factors & System Requirements

Before buying any computing device, answer these five questions systematically.

Table 4-1: Factors to Consider When Buying a Computer (Campbell et al., 2023)

Platform	Q: Do I need software that requires a specific operating system (Windows, macOS, Linux, Android)? Does this device need to be compatible with equipment I already own? ▶ *Most professional software runs on Windows. Server/development work typically requires Linux. Apple devices lock you into macOS/iOS. Match the platform to your software needs.*	**KEY TIP** *Most professional software runs on Windows.*
Hardware	Q: Do I require specific hardware components to perform my intended tasks? How much data and information do I plan to store on the device? ▶ *A video editor needs a dedicated GPU and large, fast storage. A programmer mainly needs fast CPU and lots of RAM. A student may only need basic office capability.*	**KEY TIP** *A video editor needs a dedicated GPU and large, fast storage.*
Hardware Specs	Q: Will the tasks I perform or software I want to run require certain hardware specifications — CPU speed, RAM capacity, GPU, storage type? ▶ *Always buy to the RECOMMENDED specification, not the minimum. Minimum specs produce frustrating performance. Buy the highest specs your budget allows.*	**KEY TIP** *Always buy to the RECOMMENDED specification, not the minimum.*
Form Factor	Q: Will I use this computer in one fixed location (home, office), or will I need mobility? What screen size and keyboard size suits my primary use? ▶ *Desktop: best performance per kwacha. Laptop: portability at a cost premium. Tablet: ideal for content consumption and fieldwork. All-in-one: saves desk space.*	**KEY TIP** *Desktop: best performance per kwacha.*
Add-on Devices	Q: What additional peripheral devices will I need to complete my tasks: printer, scanner, second monitor, webcam, microphone, drawing tablet? ▶ *Check port availability before buying. Many ultra-thin laptops lack USB-A and Ethernet ports. Budget for required accessories upfront to avoid unexpected costs.*	**KEY TIP** *Check port availability before buying.*

Evaluating System Requirements — Selecting the Right Specs

Principle: When choosing a computer to run multiple programs or apps, evaluate each specification requirement across ALL the software you intend to use — then select hardware that meets or exceeds the highest requirement in each category. (Campbell et al., 2023)

Table 4-2: Evaluating System Requirements (Campbell et al., 2023)

Specification	Recommended Solution
Different processor requirements	Identify the program or app with the GREATER processor requirement and select a computer with a processor that meets or exceeds that requirement. ▶ *Example: App A requires Intel Core i3; App B requires Core i5. Select Core i5 or higher — it handles both.*
Different memory requirements	Identify the program or app with the greater memory requirement and select a computer with a memory type and capacity that meets or exceeds this requirement. ▶ *Computers with as little as 4 GB are great for basic web browsing and productivity. Computers with as much as 32 GB are used for virtual reality, high-end gaming, and other intensive tasks.*
Different storage requirements	ADD the storage requirements for each program or app you want to use, and select a computer with a storage capacity that EXCEEDS the sum of all storage requirements. ▶ *Example: OS (25 GB) + App A (10 GB) + App B (15 GB) + personal files (50 GB) = 100 GB total minimum. Select at least 256 GB to allow room to grow.*
Other differing hardware requirements	In most cases, identify the program or app with the greater requirement and select a computer that at least meets or exceeds this requirement. ▶ *Examples: GPU for video editing / gaming; Bluetooth version; specific port types (Thunderbolt, USB-C); screen resolution; biometric reader.*

Which Type of Computer is Right For You? — Form Factors

Form Factor: The shape and size of the computer. Determines portability, upgradeability, performance potential, and intended use. Various types exist: desktops, all-in-ones, laptops, tablets, and other mobile devices. (Campbell et al., 2023)

🖥 Desktop Computer

Typically consists of a separate system unit, monitor, keyboard, and mouse. Remains in a stationary location under a desk or table.

✓ Best performance per kwacha — easiest to upgrade ✓ Larger components = better cooling = longer sustained performance
✗ Not portable — fixed location

📍 *Zambia: Standard in Zambian offices, schools, and CBU computer labs. Best value for a fixed workstation.*

💻 Laptop Computer

A compact communicating device with built-in input devices (keyboard, touchpad, webcam), output devices (screen, speakers), and one or more storage devices.
✓ Portable — use anywhere: lecture hall, café, field ✓ Battery-powered — works during load-shedding
✗ Higher cost per spec than desktop

📍 *Zambia: Most common student device. Ultrathin laptops: lighter, longer battery, thinner — but less powerful and more expensive.*

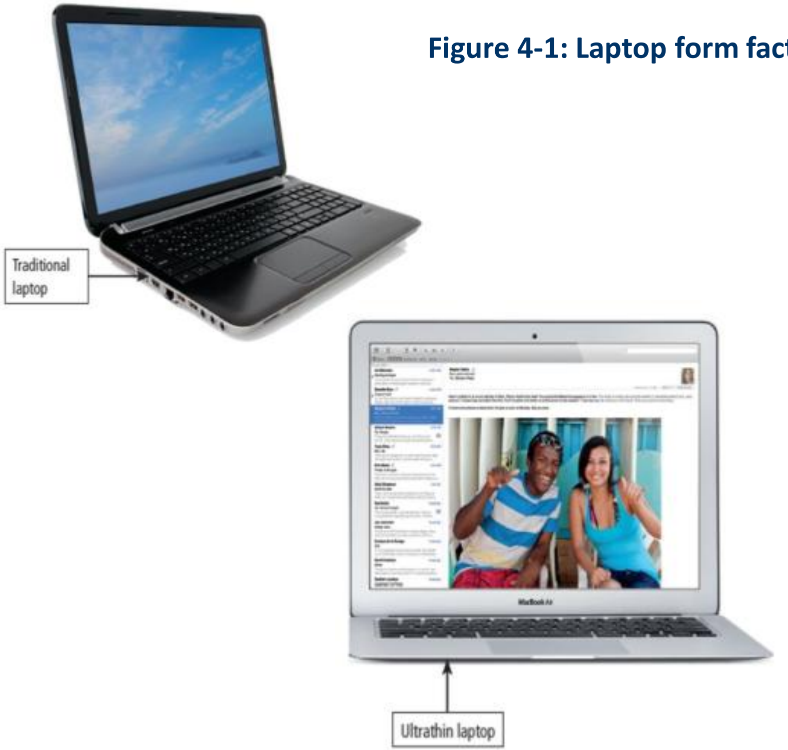


Figure 4-1: Laptop form factors

Which Type of Computer is Right For You? — Form Factors

Form Factor: The shape and size of the computer. Determines portability, upgradeability, performance potential, and intended use. Various types exist: desktops, all-in-ones, laptops, tablets, and other mobile devices. (Campbell et al., 2023)

Figure 4-2: Slate and convertible tablets

🖥 All-in-One Computer

Monitor and system unit are housed together in a single integrated unit — a sleek, space-saving design popular in reception areas and modern offices.

✓ Single device, minimal cables — clean desk setup ✓ Space saving — ideal for small offices and reception desks

✗ Not upgradeable — components are integrated

📍 *Zambia: Used in Zambian bank branches, hotel receptions, and some clinics for their clean appearance.*

Tablet

Two popular form factors: Slate (letter-sized pad without physical keyboard) and Convertible (screen on lid + keyboard on base — flips between tablet and laptop modes).

✓ Ultra-portable, lightweight, longest battery life ✓ Touch interface — ideal for field data collection

✗ Limited software — not ideal for programming

📍 *Zambia: Used by NGOs for DHIS2 data entry in rural clinics, by Zampost delivery tracking, and in some primary schools for digital learning.*

MODULE 4 · SECTION B

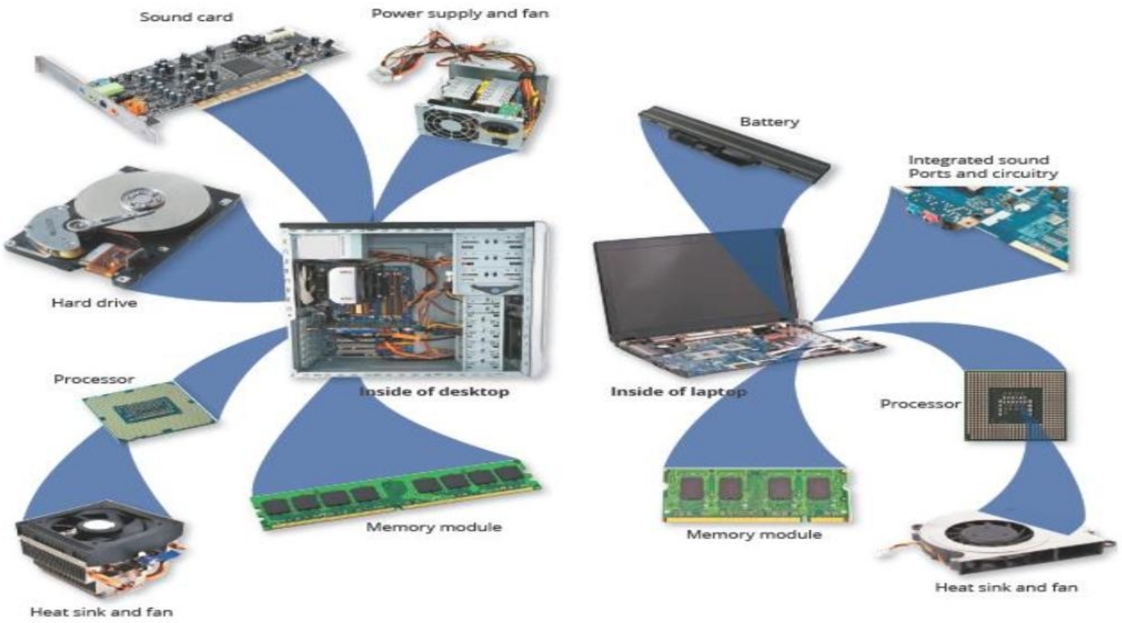
Inside the Case — Motherboard, CPU & Memory

Motherboard · Processor cores · RAM vs ROM · RAM types · Virtual memory · Machine cycle · Registers

Inside the Case

Key Principle: Processor chips for laptops, desktops, and servers can generate quite a bit of heat, which could cause the chip to malfunction or fail. Heat sinks, liquid cooling technologies, and cooling pads are used to help further dissipate processor heat. (Campbell et al., 2023)

Figure 4-3: Typical components of a higher-end desktop and laptop



Key Concepts

Heat Sink:

A small ceramic or metal component with fins on its surface that absorbs and disperses heat produced by the processor.

Liquid Cooling:

Uses a continuous flow of fluid, such as water and glycol, in a process that transfers the heated fluid away from the processor.

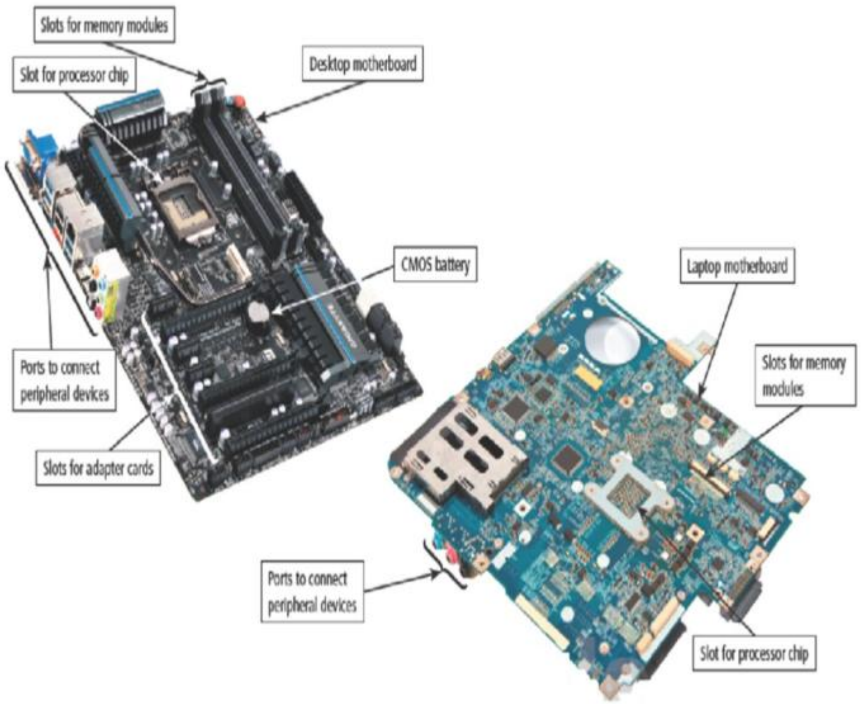
Cooling Pad:

A cooling pad rests below a laptop and protects the computer from overheating.

The Motherboard

Motherboard: A circuit board inside a computer that contains the microprocessor, computer memory, and other internal devices. Also called the main circuit board or system board. Every other component connects to or sits on the motherboard. (Campbell et al., 2023)

Figure 4-4: Typical desktop and laptop motherboard



Key Motherboard Concepts

Computer Chip:
On desktop and laptop computers, the circuitry for the processor, memory, and other components resides on a computer chip. A small piece of semiconducting material (usually silicon) on which integrated circuits are etched. Contains microscopic pathways carrying electrical current.

Integrated Circuit:
Contains many microscopic pathways capable of carrying electrical current — millions to billions of transistors on a single chip.

BIOS / UEFI:
Basic Input/Output System (or modern UEFI). First code that runs on power-on: tests hardware (POST), initialises components, loads the OS.

Expansion Slots:
PCIe x16 for GPU; PCIe x1 for sound/network cards; M.2 slots for fast NVMe SSDs. Allow adding capability without replacing the whole system.

The Processor — CPU

CPU Key Concepts

Processor Core:
A unit on the processor chip with the circuitry necessary to execute one independent stream of instructions. A quad-core CPU has 4 independent processing units sharing one physical chip.

Multi-core Processor:
A processor containing multiple cores. Processors with more cores perform better for tasks that can be parallelised — running multiple apps, video encoding, compiling code. Modern CPUs: 4–24 cores.

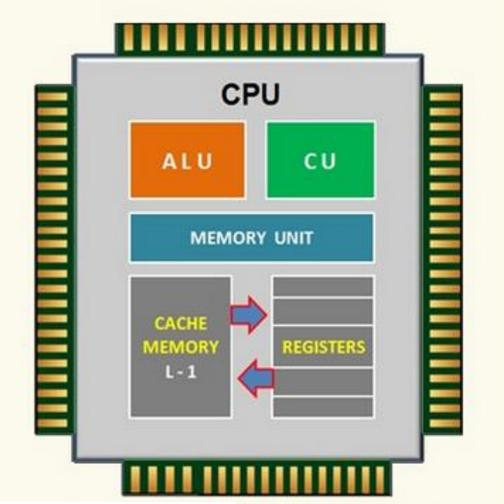
Clock Speed:
Measures how many instruction cycles per second the CPU can execute, expressed in GHz. A 3.5 GHz CPU performs 3,500,000,000 clock cycles per second.

Cycle:
The smallest unit of time a processor can measure — one pulse of the system clock. Multiple cycles are required to complete a single instruction. Efficiency is measured by Instructions Per Cycle (IPC).

Bus Width (Word Size):
Determines the speed at which data travels between the CPU, memory, and other components. Also called 'word size.' All modern CPUs are 64-bit — they process 64 binary bits per cycle.

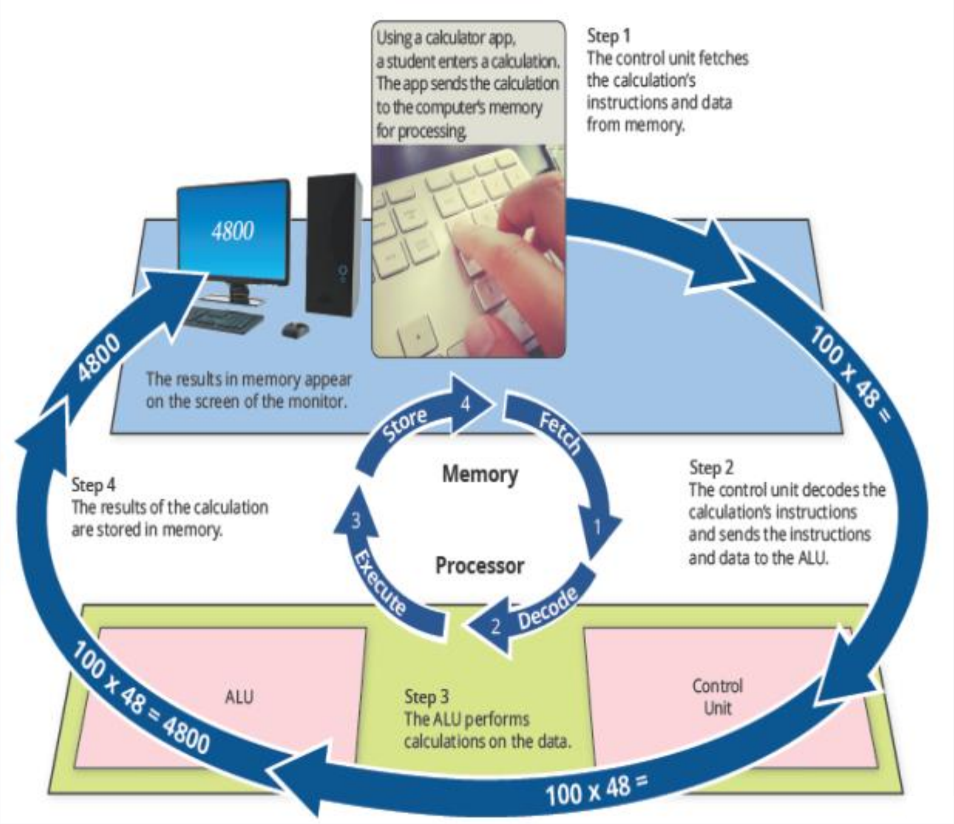
Benchmark:
A standardised test run by an independent laboratory to determine processor speed and compare performance objectively. Common tools: Cinebench R23 (CPU rendering), PassMark, SPECint.

Figure 4-5: Central Processing Unit



The Processor

Figure 4-6: The steps in a machine cycle



The Machine Cycle

A processor repeats a set of four basic operations comprising the machine cycle for every instruction it executes. (Campbell et al., 2023)

1 FETCH

The process of obtaining a program or an application instruction or data item from memory.

2 DECODE

Refers to the process of translating the instructions into signals the computer can execute.

3 EXECUTE

The process of carrying out the commands.

4 STORE

Storing, in this context, “writing” means writing the result to memory (not to a storage medium).

Computer Memory — RAM vs ROM

Computer memory is responsible for holding data and programs as they are being processed by the CPU.

RAM — Random Access Memory

Definition:	Temporary (working) storage location. Holds the OS, active applications, and all data currently being processed.
Location:	Stored on one or more chips connected to the main circuit board (the motherboard).
Volatile?:	**YES — volatile memory. Contents are LOST immediately when power is removed.**
Speed:	Much faster than any storage device. CPU accesses RAM in nanoseconds (~60ns), compared to SSD (~100 microseconds).
Capacity:	Measured in GB. Typical range: 4GB (basic) to 64GB+ (professional/gaming). Windows 11 minimum: 4GB; recommended: 8GB+.
Analogy:	*RAM is your desk — it holds everything you are actively working on. More RAM = larger desk = more tasks open simultaneously.*

ROM — Read-Only Memory

Definition:	Permanent storage location. Contains the BIOS/UEFI — the first instructions that run when the computer is powered on.
Location:	Stored on a chip containing BIOS connected to the motherboard. Cannot be written by regular applications.
Volatile?:	**NO — non-volatile memory. Contents are RETAINED when power is removed.**
Firmware:	Computer manufacturers periodically update the instructions on the ROM chip. These updates are called firmware updates. (Campbell et al., 2023)
BIOS Function:	Tests hardware (Power-On Self Test), initialises all components, and passes control to the OS bootloader on the storage drive.
Analogy:	*ROM is the printed manual — permanent, read frequently, rarely changed, essential for initial startup of the machine.*

Computer Memory — Types of RAM

Not all RAM is identical — different types serve different purposes and have different performance characteristics.

Table 4-3: Types of RAM

Type	Full Name	How It Works	Volatile or Non-Volatile	Primary Use
DRAM	Dynamic RAM	Memory needs to be constantly recharged or contents will be erased. (Campbell et al., 2023)	Volatile	Main system RAM in desktops and laptops (DDR4, DDR5 are both DRAM types)
SRAM	Static RAM	Memory can be recharged less frequently than DRAM. Faster and more expensive than DRAM. (Campbell et al., 2023)	Volatile	CPU cache memory (L1, L2, L3) — ultra-fast buffer between CPU and main RAM
MRAM	Magneto-resistive RAM	Uses magnetic charges to store contents. Can retain its contents in the absence of power. (Campbell et al., 2023)	Non-volatile	Specialist embedded and IoT systems — emerging technology
Flash	Flash Memory	Fast memory type, typically less expensive than some other RAM types. Can retain its contents without power. (Campbell et al., 2023)	Non-volatile	USB drives, SSDs, SD cards, smartphones, tablets, BIOS chips

Computer Memory — Virtual Memory

Virtual memory is a technique where the operating system uses part of a storage drive as temporary overflow space when RAM is not enough.(Campbell et al., 2023)

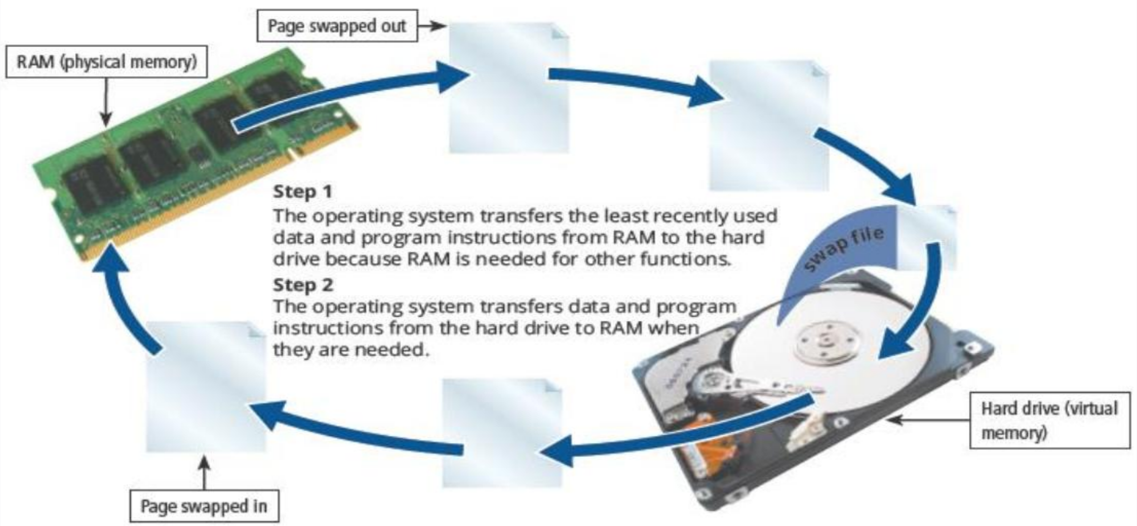


Figure 4-7: How a computer might use virtual memory

1. RAM Fills Up

An operating system (OS) and other apps require a certain amount of RAM to function properly. OS, applications, and active data consume all available RAM.

2. Swap File Used

When more apps run simultaneously, more RAM will be required. OS reserves an area of the HDD/SSD (swap file / paging file) as a temporary overflow extension of RAM.

3. Paging Occurs

The area of the hard drive temporarily used to store data that cannot fit in RAM is called a swap file, or paging file. Inactive RAM contents are moved to the swap file. Needed data is swapped back into RAM from the swap file.

4. Performance Drops

Using virtual memory may decrease your computer's performance. Disk access is 100–1000× slower than RAM. Heavy paging causes 'thrashing' — the system becomes nearly unusable.

Registers

Registers: Small, high-speed storage locations inside the CPU that temporarily hold data and instructions. Registers are part of the processor — NOT part of main memory or permanent storage. (Campbell et al., 2023)

Register Functions — What Registers Store and Do:

1

Instruction Address Register (Program Counter)

Stores the LOCATION (memory address) from which the next instruction will be fetched. After each fetch, it automatically increments to point to the next instruction in sequence.

2

Instruction Register

Stores the CURRENT INSTRUCTION while the Control Unit decodes it. Holds the binary-encoded operation code (opcode) and operands being decoded and prepared for execution.

3

Data Register (Accumulator)

Stores the DATA while the ALU (Arithmetic Logic Unit) calculates it. Holds operands (numbers being operated on) and intermediate results during arithmetic and logic operations.

4

Status / Flags Register

Stores the RESULTS of a calculation — flag bits that indicate outcomes: zero flag (result = 0), carry flag (arithmetic overflow occurred), sign flag (result is negative), overflow flag.

Memory Speed Hierarchy (fastest → slowest): Registers (~0.3ns) → L1 Cache (~1ns) → L2 Cache (~4ns) → L3 Cache (~10ns) → RAM (~60ns) → SSD (~100µs) → HDD (~10ms)

MODULE 4 · SECTION C

Input and Output Devices

Scanners · Printer types (6) · Voice synthesiser · Projectors · Installing hardware

Input Devices

An input device allows a user to enter data, commands, or instructions into a computer or mobile device.

⌨ Keyboard

Primary text input. Membrane (quiet, cheap), mechanical (tactile, durable), or virtual (touchscreen on phones/tablets). Standard on every desktop and laptop.

🖱 Mouse / Touchpad

Pointing device. Mouse: optical, laser, or trackball. Touchpad: capacitive surface built into laptops. Multi-touch gestures (pinch, scroll) on modern touchpads.

👆 Touchscreen

Capacitive touch input — detects multiple finger touches simultaneously. Built into all smartphones and tablets. Also found on modern laptops and all-in-ones.

🎙 Microphone

Converts sound waves to digital audio data. Used for voice commands (Siri, Cortana), video calls, voice recording, and speech recognition.

📷 Digital Camera / Webcam

Captures still images and video. Webcams built into laptops enable video conferencing (Zoom, Teams). External cameras connect via USB.

Barcode / QR Scanner

Optical reader converts printed codes to data. Used in Zambian supermarkets, logistics, and MTN MoMo QR payments for instant transfer authorisation.

Scanners — Input Devices That Convert Physical Documents to Digital Files (Campbell et al., 2023)

- A scanner is an input device that converts an existing physical (paper) image or document into an electronic file you can open on your computer.
- 3-D scanners can digitise three-dimensional physical objects — used in healthcare (prosthetics), manufacturing quality control, and cultural heritage preservation.
- Scanners can digitise a printed document so you can edit it in a word processor — without retyping. Requires OCR (Optical Character Recognition) software to make text editable.

Output Devices

An output device communicates processed results FROM the computer TO users, paper, or other devices.

Common Output Devices

Monitor / Screen:
Primary visual output. LCD/LED (most common), OLED (high contrast smartphones/premium laptops), and projectors for large audiences.

Voice Synthesiser:
Converts text to speech (TTS). Some apps and operating systems have built-in voice synthesisers — convenient for everyone and ESSENTIAL for visually impaired users.

Projector:
Displays output from a computer on a large surface (wall or screen). Wired or wirelessly connected. Shows an EXACT replica of the computer monitor — identical output, enlarged.

Speakers / Headphones:
Audio output — converts digital audio signals to sound waves. Built into laptops; external speakers/headphones for better quality. Critical for accessibility and multimedia.

Figure 4-8: Projector and screen

Output Devices — Printers

A printer creates hard copy output on paper, film, and other media.

Table 4-4: Types of Printers

Type	Description
Inkjet	Prints by spraying small dots of ink onto paper. Good for colour photos at home. Low purchase cost but high ink cost per page.
Laser	Uses a laser beam and toner powder to print. Fast, high-quality text. Best for high-volume office printing — lower cost per page.
Multifunction (MFD/All-in-One)	Serves as both input (copying, scanning, faxing) and output (printing) device. Space-saving for offices. Also called all-in-one printer.
Mobile Printer	Small, lightweight printer built into or attached to a mobile device. Used by delivery staff, field agents, and receipt printing on the go.
Plotter	Large-format printer using charged wires to produce high-quality drawings on large paper rolls. Used for architectural blueprints and engineering drawings.
3-D Printer	Creates physical 3D objects from digital models using layered plastics and other materials. Used in healthcare (prosthetics), manufacturing, and prototyping.

Printer Connection: A printer can be linked to a computer wirelessly (Wi-Fi), over a network (shared), or with a USB cable. Wireless printers typically require a device driver to be installed. Plug-and-play devices begin functioning as soon as they are connected — no manual driver installation needed. (Campbell et al., 2023)

Installing & Evaluating Hardware

First-Time Hardware Setup (Campbell et al., 2023)

1 **1. Check completeness**
Verify all necessary components are included and free from damage before turning on. Check for transit damage — especially to screen.

2 **2. Read the manual**
Review the setup documentation to ensure you follow the correct installation sequence. Avoid common mistakes that void warranty.

3 **3. Unpack carefully**
For a laptop: charge the battery completely before first use. This prevents battery calibration issues that affect long-term battery health.

4 **4. Connect all accessories**
Connect monitor, keyboard, mouse, and network cable before first boot. Install in the correct order per manufacturer instructions.

5 **5. Power on and configure**
Turn on and follow all on-screen setup steps. Activate Windows with your licence key. Run Windows Update fully before installing applications.

6 **6. Install device drivers**
A device driver is a program that allows your computer to issue controls to a separate device, such as a printer, monitor, or GPU. Always download from the MANUFACTURER's website — never third-party driver sites.

7 **7. Set up backup**
Configure OneDrive auto-sync for Documents folder. Enable Windows auto-save in Office apps. Set up an external drive for regular backups.

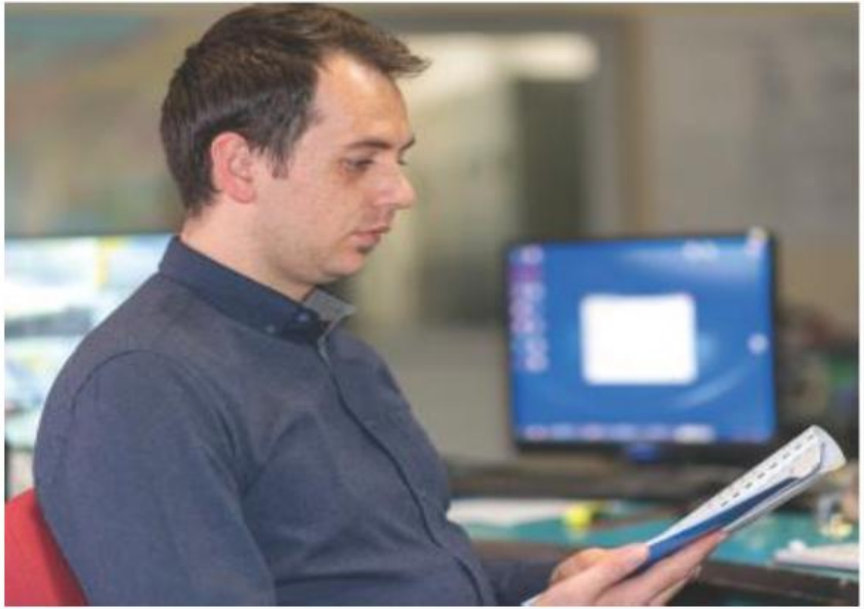


Figure 4-9: User manuals provide safe setup instructions

Installing & Evaluating Hardware

Evaluating Performance

Clock Speed:
Measures the speed at which the processor can execute instructions. Expressed in GHz. A higher clock speed means faster instruction processing per cycle.

Cycle:
The smallest unit of time a processor can measure. Each clock pulse = 1 cycle. Multiple cycles may be needed per instruction.

IPC:
Instructions Per Cycle — the efficiency metric. Modern CPUs complete 3–5 instructions per cycle using pipelining and out-of-order execution.

Bus Width:
Determines the speed at which data travels between CPU, RAM, and peripherals. Also called word size. All modern CPUs are 64-bit.

Benchmark:
A test run by a lab or organisation to determine processor speed and performance. Allows objective comparison between different hardware configurations.

Plug-and-Play Devices

Plug-and-Play (PnP): Peripheral devices that begin functioning immediately when connected — no manual driver installation required.

Examples: USB keyboards, mice, most USB drives, standard webcams.

Wireless devices: install the software driver FIRST, then connect the hardware — exact opposite of wired PnP.

Non-PnP devices (older printers, audio interfaces) require manual driver installation from manufacturer website.

MODULE 4 · SECTION D

Storage Solutions — Internal, External & Cloud

HDD vs SSD · Optical media · USB flash drives · Cloud storage · Choosing solutions

Storage Solutions — Internal, External & Cloud-Based

Storage holds data PERMANENTLY — it survives when power is removed (unlike volatile RAM). Three storage environments exist.

Internal Storage

HDD — Hard Disk Drive

The most common storage medium. Stores data magnetically on spinning platters using a read/write head. Capacities: 500GB–18TB. Slower (~100MB/s) but cheap per GB. Fragile: moving parts sensitive to drops.

SSD — Solid-State Drive

Faster and more durable than magnetic HDD drives. Uses flash memory chips with no moving parts. Speeds: 450–7,000MB/s. Higher cost per GB than HDD. Recommended as OS drive.

NVMe M.2 SSD

Connects directly to the motherboard via PCIe. Fastest consumer storage: up to 7,000MB/s. Standard in modern premium laptops. Best performance — ideal for OS and active projects.

External Storage

External Hard Drive

Portable HDD connected via USB 3.0/USB-C. Capacities: 500GB–20TB. Excellent for backups and large file transport.

USB Flash Drive

Compact solid-state storage. Capacities: 4GB–2TB. No power required. Ubiquitous for file transfer. SECURITY RISK: common malware vector — always scan before opening files.

Optical Media

CDs (700MB), DVDs (4.7GB), Blu-ray (25–100GB). Use laser technology for storage and playback. Once widely used to distribute software. Now largely replaced by USB drives, digital downloads, and cloud storage for most use cases.

Figure 4-10: Hard disk drive

Storage Solutions — Internal, External & Cloud-Based

Storage holds data PERMANENTLY — it survives when power is removed (unlike volatile RAM). Three storage environments exist.

Cloud-Based Storage

What it is

Cloud storage involves storing electronic files on a remote server connected to the Internet, not on a local computer, and is called storing data on the cloud. (Campbell et al., 2023)

It enables the storage of files remotely on servers that could be any part of the world.

Storing files on and retrieving files from cloud storage requires only a computer or mobile device with an Internet connection.

How it works

Cloud storage companies host and maintain servers anywhere in the world. Storing and retrieving files requires ONLY a device with an internet connection. Companies manage all server maintenance.

Figure 4-11: Cloud storage

MODULE 4 · SECTION E

Network Hardware — Connecting Computers

Nodes · Hub · Switch · Router · Modem · NIC · Wi-Fi security risks

Network Hardware — Components and Their Roles

All networks share the same basic hardware. Networks allow computers to share resources — hardware, software, data, and information. A network requires a combination of hardware and software to operate. Smaller networks use simpler hardware; larger networks require more sophisticated equipment.

Figure 4-12: Modems can connect many devices to the internet

Nodes	Devices on a network including computers, tablets, phones, printers, game consoles, and smart home devices. Any device that can communicate on the network.
Hub	A hub is a device that provides a central point for cables in a network and transfers all data to all devices.
Switch	A switch is similar to a hub in that it provides a central point for cables in a network.
Router	A router is a device that connects two or more networks and directs, or routes, the flow of information along the networks.
Modem	A modem is a communications device that connects a communications channel, such as the Internet, to a sending or receiving device, such as a computer.
NIC	Network Interface Card — gives a computer the hardware ability to connect to a network. Every networked device has a NIC, either built-in or as an expansion card.

Network Hardware — Components and Their Roles

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Figure 4-12: Modems can connect many devices to the internet

- Most of today's modems are digital, which means that they send and receive data to and from a digital line.
- Cable and DSL (digital subscriber line) are two common types of digital modems.
- The type of modem required for your network will depend on your Internet service provider.
- A cable modem sends and receives digital data over a cable TV connection.
- A DSL modem uses existing standard copper telephone wiring to send and receive digital data.

Wi-Fi Networks — Security Risks & Countermeasures

Wi-Fi networks are vulnerable to multiple security risks. Wireless transmissions travel through the air and can be intercepted by anyone within range.

Wi-Fi Security Risks

✗ Reading wireless transmissions:
Without encryption, Wi-Fi packets travel as readable plaintext. Anyone with Wireshark (free packet analyser) can capture usernames, passwords, and browsing content on unencrypted public Wi-Fi.

✗ Viewing or stealing computer data:
Connecting to a malicious 'evil twin' hotspot (same name as a real network) routes all your traffic through the attacker's device. Shared folder access on the network can expose local files.

✗ Injecting malware:
Attackers on the same Wi-Fi network can attempt to inject malware into unencrypted traffic or exploit network-shared resources. Man-in-the-middle attacks alter data in transit.

✗ Downloading harmful content:
Unauthorised users piggybacking your Wi-Fi network to download illegal content do so using YOUR IP address — making you legally liable for their activity under Zambia's Cyber Crimes Act 2021.

Security Best Practices

- ✓ Always use WPA3 or WPA2 encryption — NEVER WEP (broken since 2001). Check your router admin page to verify encryption type.
- ✓ Change the default router admin password immediately — default passwords are publicly listed for every router model online.
- ✓ Use a strong, unique Wi-Fi password: 12+ characters mixing letters, numbers, and symbols. Never share publicly.
- ✓ Use a VPN on ALL public Wi-Fi (airports, cafés, hotels) — encrypts all traffic so interception produces only gibberish. ProtonVPN is free.
- ✓ Disable WPS (Wi-Fi Protected Setup) on your router — it has known brute-force vulnerabilities despite appearing convenient.

MODULE 4 · SECTION F

Hardware Security, Health & Responsible Disposal

Theft prevention · Power protection · RSI & ergonomics · Cyberbullying · E-waste & SEM

Protecting Hardware — Theft, Physical Damage & Power Threats

Hardware theft, physical damage, and electrical hazards are significant threats. Proactive measures are far cheaper than replacement.

Theft Prevention

Cable Lock:

Use a Kensington cable lock to physically secure laptops to desks or fixed furniture.

Mobile Device Awareness:
Keep mobile devices out of sight when travelling in high-crime areas. Minimise use of headphones in public — they reduce situational awareness.

Tracking & Remote Wipe:

Enable 'Find My Device' (Windows) or 'Find My' (Apple). Can remotely locate, lock, and WIPE a stolen device before data is accessed.

Data Backup:
Always maintain backups (OneDrive, Google Drive, external drive) so that even if the device is stolen or destroyed, your data survives completely.

Physical & Environmental

Cleaning:
Use a damp cloth to gently clean screens. Use compressed air to clean keyboards of dust and debris. Clean laptop vents every 3–6 months to prevent overheating.

Temperature & Humidity:
Extreme temperatures and humidity damage electronics. Never leave a laptop on a car dashboard in summer (can reach 60°C+). Store in cool, dry locations.

Drop Protection:
Use padded laptop bags. HDDs are fragile — dropping a powered-on HDD can instantly destroy the platter and cause permanent data loss. SSDs are drop-proof.

Processor Cooling:
Heat sinks, liquid cooling, and cooling pads prevent CPU overheating and failure. Use cooling pads for extended laptop sessions. Never place laptop on soft surfaces blocking vents.

Power Protection

UPS (Uninterruptible Power Supply):

Maintains power to equipment during an interruption in the primary electrical source.

Surge Suppressor:
Prevents voltage spikes and surges from reaching and damaging electronic components. A lightning strike without a surge suppressor can instantly destroy a motherboard.

OS Restore:
S
Restoring an operating system reverts all settings to defaults or a previous version. Data backups protect against hardware failure, user error, software corruption, and natural disasters.

3-2-1 Backup Rule:
3 copies of all important data, on 2 different storage types (e.g. laptop + USB), with 1 copy offsite (cloud). Guarantees data survival even if the building burns down.

Electrical Hazards — Understanding Power Threats to Hardware

Electricity can destroy hardware instantly or degrade it over time. Understanding electrical threats enables correct protection selection.

Electrical Change	Explanation (Campbell et al., 2023)	Duration	Hardware Risk	Protection
Blackout	Total loss of power	Minutes to hours	Unsaved RAM data lost permanently; HDD may corrupt during active write	UPS: provides 15–45 min battery backup for safe shutdown
Brownout	Drop in voltage lasting minutes or hours	Minutes to hours	Components run below spec voltage; long-term degradation of PSU and motherboard	UPS with AVR (Automatic Voltage Regulator) stabilises supply
Spike	Very short duration of voltage increase	Microseconds	Can instantly burn out capacitors and damage CPU, GPU, and RAM	Surge suppressor power strip with spike protection
Surge	Short duration of voltage increase	Milliseconds	Can destroy motherboard traces, PSU, and all connected components	Surge suppressor; unplug during storms; whole-home lightning arrester
Noise	Unwanted high-frequency energy	Continuous	Causes computational errors, intermittent crashes, data corruption	Power conditioner; UPS with line conditioning; proper earthing

Repetitive Strain Injury

RSI (Repetitive Strain Injury): Injuries to muscles, nerves, tendons, and ligaments caused by repeated and long-term use of technology devices. Primarily affects the upper body: elbows, forearms, hands, neck, shoulders, and wrists. (Campbell et al., 2023)

Table 4-5: Causes of RSI

Cause	Description	Example
Repetitive Activity	Repeating the same motion over a lengthy time period	Typing on a keyboard for multiple hours every day, year after year
Improper Technique	Using the wrong procedure or posture during device use	Slouching in a chair; wrists bent upward while typing; head tilted to phone
Uninterrupted Intensity	Performing the same high-level activity without adequate rest breaks	Programming all day at a computer with no movement breaks

RSI Symptoms (Campbell et al., 2023)

Aching	Pain	Throbbing
Cramp	Stiffness	Tingling
Numbness	Tendernes s	Weakness

Ergonomics & Protecting Your Health

Ergonomics: *applied science specifying the design and arrangement of items you interact with so that the interaction is efficient and safe. (Campbell et al., 2023)*

Ergonomics — Prevention

Monitor height:
Screen top at or just below eye level. Neck should be neutral — no tilt forward or backward.

Keyboard & wrists:
Wrists flat or angled slightly downward. Elbows at ~90°. Avoid wrist rests during active typing.

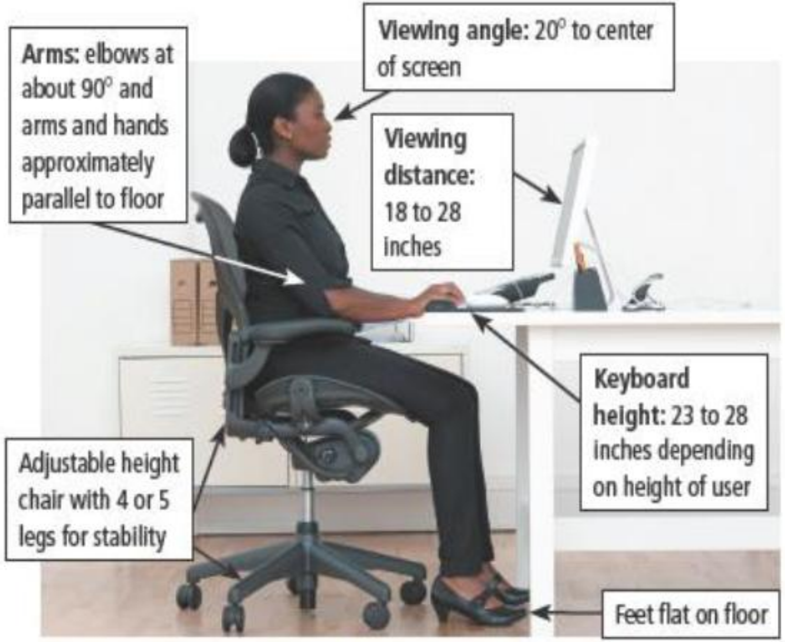
Chair & back:
Back straight, supported by lumbar region. Feet flat on floor. Thighs parallel to floor.

Viewing distance:
50–70 cm (arm's length) from eyes to screen. Prevents Computer Vision Syndrome (CVS).

Break schedule:
20-20-20 rule for eyes: every 20 min, look 20 feet away for 20 seconds. Take 5-min break every 45–60 min.

Lighting:
No glare on screen. Position screen away from bright windows. Use anti-glare screen filter if needed.

Figure 4-13: Ergonomics sitting setup



Behavioural Risks — Cyberbullying & Technology Addiction

Technology creates new forms of harm beyond physical injury. Technology addiction, cyberbullying, and cyber-stalking are serious risks with real legal consequences in Zambia.

Behavioural Risks of Technology:

Technology Addiction:	Occurs when a user is obsessed with using a technology device. Symptoms: inability to stop, irritability without device, declining academic performance, neglecting real-world responsibilities and relationships.
Sedentary Lifestyle:	Excessive screen time reduces physical activity. Linked to cardiovascular disease, obesity, and type 2 diabetes. CS students who sit for extended programming sessions without movement are at elevated risk.
Psychological Development:	Heavy social media use is linked to increased anxiety, depression, and distorted self-image, particularly in young people. Algorithmic echo chambers limit exposure to diverse perspectives.
Social Interaction:	Technology can substitute for rather than supplement real human connection. Over-reliance on digital communication can erode face-to-face social skills and emotional intelligence.

Harmful Features of Cyberbullying

Feature	Traditional Bullying	Cyberbullying
Seems to never end	Child may be bullied at school but the bullying ceases once they go home.	Because cyberbullying comments posted online are visible ALL THE TIME to the victim, the bullying never ends.
Everyone knows about it	Mean-spirited words may be heard only by those physically nearby.	A cyberbully can post comments online that can be read by EVERYONE — unlimited audience.
May follow for a lifetime	Bullying usually stops when the person leaves that environment.	Posted cyberbullying comments may remain visible ONLINE FOR YEARS — impacting college admissions and employment prospects.

E-Waste & Responsible Hardware Disposal — SEM Framework

E-Waste: Electronic waste from discarded digital devices. Often contains toxic metals — lead, mercury, cadmium, and arsenic — which contaminate ground and water supplies when improperly disposed of, causing serious long-term environmental and health harm. (Campbell et al., 2023)

Sustainable Electronics Management (SEM) Action Steps (Campbell et al., 2023)

Step 1

BUY GREEN

BUY GREEN: When purchasing new electronic equipment, buy only products that have been designed sustainably.

Zambia Context:

Zambia: ZEMA (Zambia Environmental Management Agency) is developing national e-waste regulations. Buying compliant products supports responsible supply chains and reduces the volume of toxic materials entering Zambia.

Step 2

DONATE

DONATE: Donate used but still functional equipment to a school, charity, or nonprofit organisation. (Campbell et al., 2023)

Zambia Context:

Zambia: Computer Aid International and local NGOs refurbish donated computers for Zambian schools. CBU itself has received donated equipment. Your old laptop can directly enable a rural student's education.

Step 3

RECYCLE

RECYCLE: Send equipment to a verified used electronics recycling centre.

Zambia Context:

Zambia: ZEMA operates e-waste collection points. Some private operators in Lusaka and Kitwe accept e-waste. AVOID informal e-waste burning near Lusaka's peri-urban areas — releases toxic fumes that cause neurological damage in children.

Tutorial Discussion Activity — Module 4

Groups of 3–4 · 15 minutes group work · Then present findings to class

1	A Zambian accounting company is equipping a new 20-person office in Kitwe. They need: workstations for 20 accounting staff (Excel, accounting software, internet), 2 shared network printers, file storage for 5TB of business records, reliable internet for all staff, and protection against load-shedding. (a) Recommend specific hardware for EACH requirement. (b) Identify all required network devices and draw a simple network diagram. (c) Justify your power protection choices given ZESCO's reliability in Copperbelt.	**System Evaluation**
2	A CBU student's 3-year-old laptop: takes 4 minutes to boot Windows 10, freezes when 3+ browser tabs are open, displays 'Low Disk Space' with only 15GB free on a 500GB drive, and gets very hot after 30 minutes of use. (a) Diagnose EACH of the four symptoms — which hardware component is responsible? (b) Recommend a specific, cost-effective upgrade for each issue.	**Hardware Diagnosis**
3	CBU IT department has 80 computers that are 7 years old (Intel Core i5-3rd gen, 4GB RAM, 500GB HDD, Windows 10). They want to replace all 80 with new machines. Some staff suggest dumping the old computers. As a CS student who has studied SEM: (a) Identify the environmental and legal problems with dumping e-waste in Zambia. (b) Propose THREE specific responsible alternatives using the SEM framework. (c) Assess whether any of these computers can be cost-effectively upgraded for continued use — specify which upgrades, estimate cost, and compare to buying new.	**E-Waste Ethics & SEM**
4	A CS graduate starts their first job as a software developer in Lusaka — programming 9 hours per day. After 3 months: wrist pain (left worse than right), chronic neck stiffness, daily headaches by 4pm. (a) Apply Table 3-6 to diagnose which of the THREE RSI causes are present in their work pattern. (b) Design a specific ergonomic workstation improvement plan (cover: monitor, keyboard, chair, lighting, distance). (c) Create a healthy technology use schedule that incorporates the 20-20-20 rule, the Pomodoro technique, and physical movement breaks.	**Health & Ergonomics**

Can You Now...? — Module 4 Learning Outcomes Self-Check

Tick only what you can explain without notes. Unticked items = study priorities for Quiz 4.

- ✓ Define hardware and list the five core categories of computer hardware components
- ✓ Explain the five factors to consider when purchasing a computer
- ✓ Describe the form factors: desktop, all-in-one, laptop, slate tablet, and convertible tablet
- ✓ Identify and describe the key components on a motherboard (CPU socket, RAM slots, PCIe, BIOS chip)
- ✓ Define processor core and multi-core processor; explain why more cores improve performance
- ✓ Describe the four steps of the machine cycle (Fetch, Decode, Execute, Store) with what happens at each
- ✓ Explain clock speed (GHz), cycles per second, IPC, bus width, and benchmarks as performance metrics
- ✓ Distinguish between RAM and ROM: volatility, purpose, location, and content
- ✓ Name and describe four types of RAM (DRAM, SRAM, MRAM, Flash)
- ✓ Explain virtual memory: what it is, when it is used, and why it slows performance
- ✓ Describe the four functions of CPU registers with specific examples of each
- ✓ Define scanner and describe its function; explain what a 3-D scanner does
- ✓ Name and describe the six types of printers with appropriate use cases
- ✓ Explain voice synthesiser and projector as output devices
- ✓ Distinguish between internal, external, and cloud-based storage with examples of each
- ✓ Compare HDD and SSD across five criteria: technology, speed, durability, capacity, and cost
- ✓ Define hub, switch, router, modem, and NIC — and explain each device's role in a network
- ✓ List four Wi-Fi security risks and describe a countermeasure for each
- ✓ Define RSI, state the three causes, and list six ergonomics principles
- ✓ Describe e-waste hazards and outline the three SEM steps

Practice Questions

Prepare these for Quiz 4 and the end-of-semester examination.

Hardware Fundamentals

(a) Define hardware. Give one example from each of the five core categories of computer hardware.
(b) List and explain ALL FIVE factors to consider when purchasing a computer .
(c) Describe the four steps of the machine cycle. What does the CPU do at EACH step?
(d) What is a computer chip? What is an integrated circuit? How do they relate to each other?

Memory & Storage

(a) Distinguish RAM from ROM across four criteria: definition, volatility, purpose, and location.
(b) Name and describe four types of RAM. Which are volatile? Which are non-volatile?
(c) Explain virtual memory. What is a swap file? Why does using virtual memory slow performance?
(d) Compare HDD and SSD. Which is faster? More durable? Cheaper per GB? When should you choose each?

I/O, Networking & Security

(a) Name and describe the SIX types of printers State the primary use for each.
(b) Explain the difference between a hub, a switch, and a router. Which is more efficient — a hub or a switch? Why?
(c) List the FOUR Wi-Fi security risks described. For each, describe ONE countermeasure.
(d) What is a device driver? Why must drivers be downloaded from the manufacturer's website, not third-party sites?

Health & Sustainability

(a) Define RSI. State the THREE causes of RSI and give a specific computing example of each.
(b) Define ergonomics. List SIX ergonomic principles for a healthy computer workstation setup.
(c) Define e-waste. Name TWO toxic substances found in computer e-waste and describe their environmental harm.
(d) Describe the THREE SEM action steps and give a Zambian example of each.